

PEARLAND, TX, USA

WMO: 722427

Lat: 29.519N

Lon: 95.242W

Elev: 44

StdP: 14.67

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.8	36.3	16.5	12.7	45.1	20.9	15.7	47.8	22.3	57.3	20.1	57.1	5.6	350	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	95.6	76.9	93.9	76.9	92.4	76.7	80.3	88.1	79.5	87.7	79.0	87.0	7.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	148.4	83.2	77.5	143.1	82.7	77.0	140.6	82.6	43.6	87.5	43.0	87.2	42.4	87.2	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	17.1	15.7	28.1	99.8	4.0	2.2	25.2	101.4	22.8	102.7	20.5	104.0	17.6	105.6		
			24.8	81.6	3.3	1.5	22.4	82.7	20.4	83.5	18.5	84.4	16.1	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.9	54.5	58.3	64.4	70.3	77.2	82.7	83.9	84.5	81.1	73.0	63.4	57.1
	DBStd	12.55	9.18	9.33	7.91	6.15	4.65	2.69	2.55	2.78	3.52	6.94	8.65	9.18
	HDD50	133	56	29	6	0	0	0	0	0	0	0	8	34
	HDD65	1132	343	224	108	25	2	0	0	0	0	18	132	280
	CDD50	7772	197	260	454	610	842	981	1052	1069	932	712	409	254
	CDD65	3293	18	35	91	184	379	531	587	604	482	265	83	34
	CDH74	33063	47	118	362	1115	3426	6078	6908	7209	4873	2312	488	127
	CDH80	13272	0	12	37	175	1055	2692	3148	3386	1978	715	69	5
(14) Wind	WSAvg	6.5	7.5	8.0	8.0	8.0	7.5	5.7	4.5	4.5	4.9	5.8	6.5	7.3
(15) Precipitation	PrecAvg	53.2	4.3	3.0	3.7	4.1	5.2	5.8	4.5	4.3	4.7	5.4	4.2	4.1
	PrecMax	79.5	10.5	9.1	8.5	9.8	13.2	24.6	12.1	9.1	11.5	17.6	12.7	8.3
	PrecMin	23.7	0.6	0.5	0.8	0.1	0.0	0.3	1.4	0.5	1.6	0.0	0.7	1.1
	PrecStd	13.0	2.4	2.2	2.1	3.0	3.3	4.8	2.9	2.6	2.3	4.5	2.8	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	81.8	83.6	86.7	91.6	97.2	97.0	98.4	95.4	91.2	85.4	80.4
		MCWB	65.4	66.4	68.6	70.2	73.8	74.6	76.5	76.8	75.4	75.2	72.8	70.3
	2%	DB	75.1	77.4	80.5	84.0	89.4	93.5	94.9	96.1	93.0	88.9	81.9	77.4
		MCWB	65.9	67.5	68.1	70.1	73.7	76.1	77.2	77.4	75.9	75.1	71.4	69.9
	5%	DB	72.5	74.5	78.0	81.9	87.4	91.8	93.2	94.1	91.2	86.5	79.4	74.8
		MCWB	65.4	67.0	67.6	70.4	73.5	76.2	77.2	77.5	75.8	73.8	69.9	68.9
	10%	DB	70.0	72.2	75.4	80.1	85.6	90.4	91.5	92.3	89.5	84.1	76.7	72.4
		MCWB	64.4	66.2	67.0	69.4	73.3	76.1	77.1	77.4	75.3	72.0	69.2	67.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	72.4	73.0	76.4	78.4	81.0	80.6	81.2	80.4	79.3	76.4	73.7
		MCDB	73.8	75.4	77.4	82.0	85.3	87.6	88.5	89.8	87.0	84.4	80.8	76.8
	2%	WB	69.3	70.9	71.6	74.8	77.5	79.8	79.9	80.2	79.3	77.9	74.4	72.1
		MCDB	72.5	74.5	76.1	80.1	83.6	86.8	88.3	89.0	86.3	84.1	78.9	75.5
	5%	WB	67.7	69.4	70.7	73.4	76.5	78.8	79.2	79.5	78.4	76.8	72.9	70.6
		MCDB	71.3	72.8	75.0	78.3	82.8	86.2	87.8	88.3	85.4	83.2	77.3	74.0
	10%	WB	65.5	67.6	69.6	72.4	75.7	78.1	78.4	78.9	77.8	75.7	71.1	68.3
		MCDB	69.2	71.2	73.8	77.2	82.1	85.8	86.9	87.6	84.8	81.8	75.5	71.7
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	17.1	17.0	16.7	15.6	15.4	15.2	15.9	15.6	18.4	18.9	17.5
		MCDBR	19.6	17.6	16.9	16.8	17.1	17.6	17.5	18.2	17.6	17.6	18.6	18.1
		MCWBR	13.1	10.7	8.7	7.7	6.1	5.2	4.9	4.9	5.4	7.1	9.9	12.2
		MCDBR	16.0	14.4	13.6	12.4	13.3	14.1	15.3	16.2	14.3	14.8	15.5	16.3
	5% WB	MCWBR	13.3	11.1	8.8	7.4	5.8	5.3	5.1	5.1	5.5	7.5	10.4	13.0
(40) Clear-Sky Solar Irradiance	taub		0.344	0.361	0.375	0.439	0.480	0.471	0.499	0.485	0.457	0.380	0.372	0.353
	taud		2.505	2.466	2.416	2.236	2.157	2.235	2.178	2.217	2.263	2.471	2.441	2.484
	Ebn at Noon		283	287	289	273	261	262	254	257	260	277	270	273
	Edh at Noon		29	32	36	45	49	45	47	45	42	32	30	28
(44) All-Sky Solar Radiation	RadAvg		916	1034	1418	1688	1888	1954	1885	1811	1573	1382	1046	816
	RadStd		88	175	119	124	134	154	182	151	168	165	106	96

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+175

Nomenclature: See separate page